

What is Node js

1. Node js is runtime environment for javaScript.
2. Node we can run JavaScript outside to the browser

Scripting Languages

Like server, windows, mac , linux etc

1. JavaScript run inside the browser.

Where we need JS outside browser

1. [Web browser](#) Run on client computer.
2. So Javascript do not handle things on server.
3. So we can not connect javascript with Database directly
4. Can not handle File system etc
5. So we have to run javascript on browser and Node help for this

Why need Node

1. Unified Language
2. High Performance
3. Good for Real-time Applications
4. Good for Data Streaming
5. Etc

How to use it full stack and FE technology

1. We make API in node js.
2. Integrate APIs with FE technology.
3. BE+FE = Full Stack.

Java (Programming Language)

History Of Node js

1. First Release :- May 27, 2009
2. Current Version: V24.21.0
3. Written in JS, C and c++

Part###

Node.js Setup on MacOS

Step 1: Download Node.js

1. Go to the official Node.js website:
2.  <https://nodejs.org/en>
3. Click on **Download Node.js (LTS)** version.
4. *(LTS means Long Term Support – best for most users.)*

Scripting Languages

Step 2: Install Node.js

1. Open the downloaded file and complete the installation steps.
2. Once installed, Node.js and **npm (Node Package Manager)** will be available.

Check Installed Versions

Open **Terminal**.

Type the following commands to verify installation:

node -v

Mac OS

```
npm -v
```

This will show you the version numbers of Node.js and npm installed.

Node.js Setup on Windows

Step 1: Download Node.js

1. Go to:
2.  <https://nodejs.org/en>
3. Click on the **Download Node.js (LTS)** version.

Step 2: Install Node.js

1. Run the downloaded setup file and complete the installation.
2. This installs both **Node.js** and **npm**.

Step 3: Check Installed Versions

1. Open **Command Prompt (CMD)**.
2. Run these commands:

Java (Programming Language)

```
node -v
```

```
npm -v
```

You should see version numbers if everything was installed correctly.

Install VS Code Editor

To write and run Node.js code smoothly, install **Visual Studio Code** (VS Code) — a lightweight code editor.

For [macOS](#):

1. Visit: <https://code.visualstudio.com/>

2. Click **Download for macOS** and follow the install instructions.

Scripting Languages

For Windows:

1. Same site: <https://code.visualstudio.com/>
2. Click **Download for Windows**.

Once installed, open VS Code and start coding in JavaScript and Node.js easily.

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Create Folder and File

Create Folder

On Desktop, create a folder manually (e.g., NodeApp)

or

Open Terminal and run:

```
mkdir NodeApp
```

```
cd NodeApp
```

Open Folder in VS Code

Using VS Code directly

Open **VS Code**

Go to **File > Open Folder**

Select your NodeApp folder

Using Terminal

code .

This opens the current folder in VS Code (if code command is configured).

Create a JavaScript File

1. Create a file named app.js

Node.js runs JavaScript files, so .js extension is required.

Scripting Languages

Write Some Basic Code

Basic Output

```
console.log(20 + 20); // Output: 40
```

Function Example

```
function fruit(item) {  
  console.log("Fruit name is " + item);  
  Java (Programming Language)  
}  
fruit("apple");
```

Variable Example

```
var a = 10;
```

```
var b = 20;
```

```
console.log(a + b); // Output: 30
```

Run Node.js Code

Steps:

1. Open terminal
2. Navigate to your project folder (if not already)
3. Run:

```
node app.js
```

File System Example (Just to Show)

```
var fs = require("fs"); // Import file system module
```

Operating Systems

```
fs.writeFileSync("anil.txt", "My name is Anil");
```

This creates a file anil.txt and writes the content inside it.

REPL (Read Eval Print Loop)

1. Just type node in terminal to enter REPL mode
2. Then type JS code line by line

```
> 2 + 2
```

```
4
```

```
> .exit
```

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What Are Modules and Their Types?

Core Modules

1. Core modules are already available in Node.js.
2. You do not need to install them.

```
const fs = require("fs");
```

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```
const os = require("os"); // import os module
```

```
fs.writeFileSync("dummy.txt", "trying with modules");
```

```
// Create a file with name and write content in it
```

```
console.log(os.platform()); // Print the platform of the OS
```

```
console.log(os.hostname()); // Print the hostname of the OS
```

```
console.log(os.cpus()); // Print the CPU information of the OS
```

Third-party Modules

You can install them from third party.

Custom Modules

You create them by yourself.

Earn a CS Degree

What Are Core Modules?

1. Core modules are modules that are already included in Node.js.
2. No need to install them separately. Just use `require()` to import them.

What Are Global Objects?

1. Global object is an object that is available in all modules by default.
2. You do not need to import or require it to use it.

You can import console forcefully:

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```
const { log } = require("console"); // import console module
```

```
log("custom log");
```

Difference Between `console.log` in JS and Node.js

The difference is:

1. In **JavaScript** → `console.log` prints on the browser console.
2. In **Node.js** → `console.log` prints on the terminal.

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Why Do We Need a Server?

JavaScript by default runs in the browser.

But when we want to run JavaScript on the server, we need Node.js.

Now, if you write some code in Node.js — how can you access that code in the browser?

It's important to understand this:

If you want to see your Node.js output in the browser, you need to set up a server.

Write Code to Create a Server

Node.js provides a built-in package called `http` to create servers.

You don't need to install it separately.

Here's the basic code to create a server:

```
const http = require("http");
```

```
http.createServer().listen(4800);
```

How to Run the Code

Open your terminal and run the file using:

```
node basics.js
```

How to Check the Output in the Browser

Open your browser and type:

```
http://localhost:4800
```

You can also check the response in:

Inspect → Network tab

Understanding Request and Response Parameters

If you want your server to return some data to the browser, this is how it works:

1. The **client** (browser) sends a request.
2. The **server** responds to that request.

In Node.js, you can handle this using the `http.createServer()` method, which accepts a callback function with request and response parameters.

`http`

```
.createServer((req, resp) => {  
  resp.write("<h1>Hello, this is Anil Sidhu</h1>");  
  resp.end("Hello"); // Ends the response  
})  
  
.listen(4800);
```

Now, when you visit `http://localhost:4800`, you will see the message in the browser.

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What is an External Package?

An external package is a package we install from a third party (someone else wrote it).

We use it to add **extra functionality** to our project without writing everything ourselves.

Example:

nodemon — a package that automatically restarts the server when code changes.

Without this, we would have to stop and start the server manually after every change.

How to Install an External Package

Example: Install the colors package from <https://www.npmjs.com/package/colors>

Open the terminal and run:

```
npm install colors
```

Try Cloud Tools

or

```
npm i colors
```

Example of Using an External Package

```
const colors = require("colors");  
  
console.log(colors.red("Hi, this is Anil Sidhu"));  
console.log(colors.green("Hi, this is Anil Sidhu"));  
console.log(colors.yellow("Hi, this is Anil Sidhu"));
```

To run this code, type in the terminal:

```
node colors.js
```

Follow Stars

What is package.json?

package.json is a file that contains details about your project, such as:

1. Project name
2. Project version
3. External packages used (dependencies)
4. Project description
5. Author
6. License

What is package-lock.json?

package-lock.json is a file that contains detailed information about the exact versions of every installed package **and its dependencies** (internal packages).

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It ensures that everyone working on the project installs **the same package versions**.

What is node_modules?

node_modules is a folder where all installed external packages are stored.

How to Create a package.json File

Run this command in the terminal:

```
npm init
```

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You will be asked to fill in some details:

package name: node-2025

version: (1.0.0)

description: node js tutorial in hindi 2025

entry point: (index.js)

git repository:

keywords:

author: Anil Sidhu

license: (ISC)

Important Notes

1. External packages are stored inside the node_modules folder.
2. **You should not push node_modules to GitHub** — because it is a very large folder.
3. Other developers can easily install the required packages by running:

Try Cloud Tools

npm install

This will read package.json and package-lock.json and install everything needed.

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Why do we need the nodemon package?

When you're working with a Node.js server, every time you make a change in your server file, you have to:

1. **Manually stop** the server.
2. **Restart** it to see the changes.

This process becomes annoying during development

Solution: nodemon

1. nodemon automatically **watches your files**.
2. It **restarts the server** automatically whenever you make changes.
3. Saves **time** and improves **developer experience**.

How to Install Nodemon

Open your terminal and run:

npm install nodemon

How to Use Nodemon

nodemon app.js

Now the server will restart automatically every time you make changes to app.js

Can we create more than one server?

Yes, we can run multiple servers in Node.js — just make sure they run on **different ports**.

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What is a Server Response?

1. When a **client** (like a browser) sends a request to a server (e.g., by visiting google.com), the **server processes the request** and sends back a **response**.
2. This response can be **HTML, JSON, plain text**, files, or anything else.

Java (Programming Language)

Example:

When you visit **google.com**, the server sends back HTML that builds the **Google homepage** on your screen.

Basic Example of Server Response

```
const http = require("http");
```

```
http
```

```
.createServer((req, resp) => {  
  resp.setHeader("Content-Type", "text/html"); // Tell browser the response is HTML  
  resp.write("<h1>Hello Anil Sidhu</h1>");  
  resp.end(); // End the response  
})  
.  
listen("4800");
```

How to run:

1. Save as app.js
2. Run: nodemon app.js or node app.js
3. Open your browser and go to: <http://localhost:4800>

Passing Variables and Functions in Response

You can also include **variables**, **functions**, or **dynamic values** (like current time):

```
const http = require("http");
```

```
const age = 29;
```

Web Design & Development

```
http
```

```
.createServer((req, resp) => {
```

```
  resp.setHeader("Content-Type", "text/html");
```

```
  resp.write(`
```

```
    <html>
```

```
      <head>
```

```
        <title>Code Step by Step</title>
```

```
      </head>
```

```
      <body>
```

```
        <h1>Hello Anil Sidhu</h1>
```

```
        <h2>Age: ${age}</h2>
```

```
        <h2>Current Date & Time: ${Date()}</h2>
```

```
      </body>
```

```
    </html>
```

```
`);
```

```
  resp.end();
```

```
})  
    .listen(4800);
```

1. `resp.setHeader("Content-Type", "text/html")`: Tells the browser that we're sending back HTML.
2. Template literals (`` ``): Used to inject JavaScript variables like `age` and `Date()` into the HTML.

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Create Server in `simpleAPI.js`

We'll use Node.js `http` module to create a basic server.

Learn to Code

Define Static API Data

```
const userData = [  
  {  
    name: "Anil",  
    age: 30,  
    email: "anil@test.com",  
  },  
  {  
    name: "sam",  
    age: 20,  
    email: "sam@test.com",  
  },  
]
```

```
{  
  name: "peter",  
  Learn Java  
  age: 30,  
  email: "peter@test.com",  
},  
];
```

This is sample static [data](#) to simulate API response.

Use This Data in Server Response

```
const http = require("http");  
  
http  
  .createServer((req, resp) => {  
    resp.setHeader("Content-Type", "application/json"); // Set the content type to JSON  
    Learn Scripting  
    resp.write(JSON.stringify(userData)); // Convert the userData array to a JSON string  
    resp.end(); // End the response  
  })  
  .listen(6000);
```

How to Test the API

Run the file using:

```
nodemon simpleAPI.js
```

Open browser and go to:

Browse Art Prints

http://localhost:6000

Test API using Thunder Client

You can also use Thunder Client (VS Code extension) or Postman:

1. Method: GET
2. URL: http://localhost:6000

You will get the JSON response.

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What is Command Line Input?

Sometimes, we want to pass values **directly while running a file** from the terminal — like a username or a dynamic port number.

Create File: command-line-input.js

```
nodemon command-line-input.js anil sidhu
```

This runs the file and passes "anil" and "sidhu" as input arguments.

How to Get Command Line Input

```
const arg = process.argv;
```

```
console.log("-----", arg);
```

Output:----- [

```
'C:\\Program Files\\nodejs\\node.exe',  
'C:\\Users\\sheet\\OneDrive\\Desktop\\tutorial\\node 2025\\command-line-input.js',  
'anil',  
'sidhu'  
]
```

□ process.argv is an array:

1. Index 0 → Node path
2. Index 1 → File path
3. Index 2 onwards → Actual input values

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Access Individual Input

```
console.log("-----", arg[3]);
```

Output:

```
----- sidhu
```

Note: Space-separated values are stored as separate elements in the array.

Example Use Case: Dynamic Port

```
const http = require("http");
```

```
const arg = process.argv;
```

```
const port = arg[2];
```

```
http
```

```
.createServer((req, resp) => {
```

```
  resp.write("test input from cmd");
```

```
    resp.end();  
  })  
  .listen(port);
```

Run this from Terminal:

nodemon command-line-input.js 4100

This will start a server

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Why load an HTML file with fs?

1. Instead of writing HTML as a string in `resp.write()`, it's more scalable and readable to keep HTML in a separate `.html` file.
2. This approach is cleaner, especially for large or structured HTML.

Computer Drives & Storage

Server File (web.js)

```
const http = require("http");
```

```
http.createServer((req, resp) => {}).listen(3200);
```

1. Creates a valid HTTP server.
2. Listens on port 3200.

HTML File (web.html)

```
<!DOCTYPE html>
```

Operating Systems

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8" />
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

```
<title>Web page</title>
```

```
</head>
```

```
<body>
```

```
<h1>Web Page Heading</h1>
```

```
<input type="text" />
```

```
<button>Click</button>
```

```
</body>
```

```
</html>
```

1. Well-formed HTML structure.
2. Proper title, meta tags, and UI elements.

Read and Load HTML File Using fs Module

```
const http = require("http");
```

Web Design & Development

```
const fs = require("fs");
```

```
http
```

```
.createServer((req, resp) => {
```

```
  fs.readFile("html/web.html", "utf-8", (err, data) => {
```

```
if (err) {  
  resp.writeHead(500, {  
    "Content-Type": "text/plain",  
  });  
  resp.write("internal server error");  
  resp.end();  
  return;  
}
```

```
resp.writeHead(200, {  
  "Content-Type": "text/html",  
});  
resp.write(data);  
resp.end();
```

Computer Drives & Storage

```
});  
}).listen(3200);
```

Correctly reads the file `html/web.html` (assuming the folder structure is correct).

1. Sets appropriate status codes (500 for error, 200 for success).
2. Handles errors gracefully.
3. Sends the HTML content as response.

Check Output

1. When you run `node web.js` or `nodemon web.js`, and visit <http://localhost:3200>, the browser will correctly show the HTML content from `web.html`.